ISEP

Segurança de Sistemas e Informação (SEGSI)
PL11
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Exercise 1: Firewalls

Firewalls, being software or device-based, are deployed in the environment to control Ingress Flows. They ensure that unsolicited flows can reach services that would otherwise be private. They help reduce the exposure of a service such that, if a new service is run, it would not be immediately exposed.

1. Investigate and describe what is IP Tables[1];
2. From your previous laboratory, try to implement IPtables on the Linux machine such that you can access TCP/22 (SSH) from your machine but deny access from the Windows Server. (If you delete those machines you can always spin up a new Linux machine); Allow all egress flows such that all domain services still work; Did RPC calls break[2]?
3. Persist the changes and try to reboot the server;
4. Take the opportunity to learn about the Windows Firewall with Advanced Security[3];
5. Create a Rule to accept TCP/3389 (RDP) from the Linux Server; test it by trying to connect directly from your machine;
6. Investigate the possibility of creating an IPSec Tunnel; In what cases do you seem fit to implement such a rule[4]?

[1] https://wiki.archlinux.org/title/Iptables
[2] https://www.digitalocean.com/community/tutorials/iptables-essentials-common-firewall-rules-and-commands
[3] https://learn.microsoft.com/en-us/windows/security/operating-system-security/network-security/windows-firewall/
[4] https://techcommunity.microsoft.com/t5/core-infrastructure-and-security/securing-dc-to-dc-communication-with-ipsec-using-windows/ba-p/257700

Exercise 2: Proxies

Proxies on the other hand try to outbound traffic. This is to block internal services from reaching malicious websites or C2 infrastructure. Modern Firewalls already include modules that act as a proxy.

1. Do the practical exercises from Exercise 1;
2. Install Squid[5] as a proxy;
3. Change the previous IPTables to block all egress traffic, except for the new installation of the proxy;
4. Change the configuration of the proxy to allow all ipp.pt domains and only www.sapo.pt [6];
5. Configure the machine such that every HTTP request uses the proxy by default.

[5] https://ubuntu.com/server/docs/proxy-servers-squid/
[6] https://www.digitalocean.com/community/tutorials/how-to-set-up-squid-proxy-on-ubuntu-20-04